SYNTHETIC SAMPLE DOCUMENT

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LINEAR REGRESSION ASSUMPTIONS

Ordinary least squares estimates coefficients by minimizing the sum of squared residuals. The estimator is unbiased when the errors have zero mean and are uncorrelated with the predictors, and it is efficient when the errors are homoscedastic and independent.

Violations matter in different ways. Heteroscedasticity leaves the coefficients unbiased but invalidates the standard errors, so inference becomes unreliable even though the point estimates look fine. Multicollinearity inflates coefficient variance without reducing predictive accuracy, which is why a model can forecast well and still be uninterpretable.

REGULARIZED REGRESSION

Ridge regression adds an L2 penalty that shrinks coefficients toward zero and stabilizes estimates under collinearity, but it never sets a coefficient exactly to zero. Lasso adds an L1 penalty and produces sparse solutions that double as feature selection, at the cost of arbitrary choices among correlated predictors. Elastic net mixes both penalties and is the pragmatic default when predictors are grouped.

CAUSAL INFERENCE

A regression coefficient is a causal effect only under strong assumptions. Difference-in-differences compares the change in a treated group against the change in an untreated group, and it requires parallel trends: absent treatment, both groups would have moved together. Instrumental variables recover a causal effect when an instrument affects the outcome only through the treatment, an exclusion restriction that cannot be tested from the data alone.

Propensity score matching estimates the effect of a binary treatment by modelling the probability of receiving it and comparing units with similar scores. It balances observed confounders only, so unmeasured confounding remains the central threat to every observational design.